Requirement Analysis

Introduction

Requirement analysis is an important phase in the development of any software project. In this phase, the needs and expectations of the users are identified and analysed. It helps developers understand what features and functions should be included in the system. Proper requirement analysis ensures that the final application solves the real problems of the users.

For the Plant Care application, requirement analysis was done to understand what plant owners need in order to take care of their plants more effectively. Many users face problems such as forgetting to water plants, lack of knowledge about plant care, and difficulty identifying plant diseases. Therefore, the application should provide features that make plant care easier and more organized.

Functional Requirements

Functional requirements describe the main features and operations that the application must perform. These functions define how the system will behave and what services it will provide to the users.

The main functional requirements of the Plant Care application are listed below:

1. User Registration and Login
The application should allow users to create an account and log in using their email or other authentication methods. This helps in storing user data and plant information securely.

2. Add and Manage Plants
Users should be able to add their plants to the application. They can enter details such as plant name, plant type, and planting date. The app should also allow users to edit or delete plant information whenever needed.

3. Watering and Care Reminders
The system should send notifications or reminders to users for watering plants, fertilizing, or other plant care activities. This feature will help users maintain a proper plant care routine.

4. Plant Information and Care Guide
The application should provide information about different plant species. This may include details such as watering frequency, sunlight requirements, soil type, and general care instructions.

5. Plant Identification Feature
Users should be able to upload a photo of a plant, and the system should help identify the plant type. This feature will be useful for users who do not know the name of their plant.

6. Disease Detection Assistance
The application can help users identify possible plant diseases by analyzing images of leaves and suggesting solutions.

7. Dashboard for Plant Overview
The application should provide a dashboard where users can see all their plants along with their upcoming care tasks.

Non-Functional Requirements

Non-functional requirements describe how the system should perform rather than what it should do. These requirements focus on performance, usability, and security.

1. User-Friendly Interface
The application should have a simple and easy-to-use interface so that even beginners can use it without difficulty.

2. Performance
The system should respond quickly when users add plants, view plant information, or receive reminders.

3. Reliability
The application should work consistently without frequent errors or crashes.

4. Security
User information such as login details and plant data should be stored securely to prevent unauthorized access.

5. Compatibility
The application should run smoothly on most Android smartphones.

Hardware Requirements

The hardware requirements describe the physical devices needed to run or develop the system.

- Android smartphone with internet connectivity
- Minimum 4GB RAM for smooth performance
- Storage space for installing the application and storing plant images

For development purposes, a computer or laptop is also required.

Software Requirements

The software requirements include the tools and technologies used to develop and run the application.

Some of the main software requirements are:

- **Android Studio** for developing the mobile application
- **Programming languages** such as Java or Kotlin
- **Firebase or database system** for storing user data
- **Internet connection** for plant identification and data synchronization
- **Operating system** such as Windows, macOS, or Linux for development

User Requirements

The Plant Care application is designed to meet the needs of users who want to maintain healthy plants but may not have enough knowledge or time to manage plant care manually.

The main user requirements include:

- Easy method to track plant care activities
- Quick access to plant information
- Reminders for watering and maintenance
- Ability to identify unknown plants
- Simple navigation and clear instructions

These requirements ensure that the application is useful for both beginners and experienced plant owners.

Summary

Requirement analysis helps in clearly understanding what the Plant Care application should do and how it should function. By identifying functional and non-functional requirements, developers can design the system more effectively.

The requirements gathered in this phase act as a guideline for the development process and help ensure that the final application meets the needs of the users. With the right features and proper design, the Plant Care app can become a helpful tool for people who want to take better care of their plants.